WORKED SOLUTION

STEP II 1992, Question 1

Pure • Limits of Sequences

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q1

The key idea

Each limit needs one standard move, and the skill is picking the right one. For rational functions, divide top and bottom by the highest power of n. For something bounded divided by something growing, squeeze it. For a known limit in disguise, substitute so the disguise falls away. And for a difference of square roots, multiply by the conjugate: that turns a difference of two nearly equal quantities into a quotient you can see clearly.

Solution

Part (i)

Dividing top and bottom by n,

$$\frac{n}{n+1} = \frac{1}{1+\frac{1}{n}} \longrightarrow \frac{1}{1+0} = 1,$$

since $\frac{1}{n} \to 0$. The limit is 1.

Part (ii)

The highest power is n^2, which appears only in the denominator. Dividing top and bottom by n^2,

$$\frac{5n+1}{n^2-3n+4} = \frac{\frac{5}{n}+\frac{1}{n^2}}{1-\frac{3}{n}+\frac{4}{n^2}} \longrightarrow \frac{0}{1} = 0.$$

The limit is 0: the denominator grows like n^2, but the numerator only like n.

Part (iii)

We know nothing useful about $\sin n$ for large integers n except that it is bounded: $|\sin n| \leq 1$. That is enough, because

$$\left|\frac{\sin n}{n}\right| \leq \frac{1}{n} \longrightarrow 0.$$

So $\dfrac{\sin n}{n}$ is squeezed between $-\frac{1}{n}$ and $\frac{1}{n}$, both of which tend to 0, and the limit is 0.

Part (iv)

? **Ask yourself** Isn't this the same as (iii)?

No: here the angle $1/n$ is shrinking to 0, not growing. Put $x = 1/n$. As $n \to \infty$ we have $x \to 0^+$, and the expression is $\dfrac{\sin x}{x}$, whose limit as $x \to 0$ is the standard result

$$\lim_{x \to 0} \frac{\sin x}{x} = 1.$$

(One way to see it: this is the derivative of sin at 0, namely $\cos 0 = 1$. Or use $\sin x = x - \frac{x^3}{6} + \cdots$, so $\frac{\sin x}{x} = 1 - \frac{x^2}{6} + \cdots$.) The limit is 1.

Part (v)

As $n \to \infty$, $\arctan n \to \frac{\pi}{2}$, because $\tan\theta \to +\infty$ as $\theta \to \frac{\pi}{2}$ from below and arctan is continuous and increasing. The limit $\frac{\pi}{2}$ is not zero, so we may take reciprocals:

$$(\arctan n)^{-1} \longrightarrow \frac{1}{\pi/2} = \frac{2}{\pi}.$$

Note that $(\arctan n)^{-1}$ means $1/\arctan n$, not the inverse function of arctan (which would just give tan).

Part (vi)

Top and bottom both tend to 0, so we can't just take limits separately. Multiply each by its conjugate:

$$\sqrt{n+1} - \sqrt{n} = \frac{(n+1) - n}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{\sqrt{n+1} + \sqrt{n}}, \qquad \sqrt{n+2} - \sqrt{n} = \frac{2}{\sqrt{n+2} + \sqrt{n}}.$$

Hence

$$\frac{\sqrt{n+1} - \sqrt{n}}{\sqrt{n+2} - \sqrt{n}} = \frac{\sqrt{n+2} + \sqrt{n}}{2(\sqrt{n+1} + \sqrt{n})} = \frac{\sqrt{1 + \frac{2}{n}} + 1}{2(\sqrt{1 + \frac{1}{n}} + 1)},$$

dividing top and bottom by $\sqrt{n}$. As $n \to \infty$ both square roots tend to 1 (since $\sqrt{\ }$ is continuous), so the limit is

$$\frac{1+1}{2(1+1)} = \frac{1}{2}.$$

Ask yourself Why should the answer be $\frac{1}{2}$ rather than, say, 1?

Near a large n the graph of $\sqrt{x}$ is almost a straight line with gradient $\frac{1}{2\sqrt{n}}$. So $\sqrt{n+1} - \sqrt{n} \approx \frac{1}{2\sqrt{n}}$ (a step of 1) and $\sqrt{n+2} - \sqrt{n} \approx \frac{2}{2\sqrt{n}}$ (a step of 2): the ratio is about $\frac{1}{2}$, the ratio of the step lengths.

Checks

Plug in a large n. With $n = 10^6$: (i) gives 0.999999; (ii) gives 5×10^{-6}; (iii) gives about -3.5×10^{-7}; (iv) gives 1.0000000 to seven places; (v) gives 0.636620, against $2/\pi \approx 0.636620$; and (vi), computed from the rationalised form, gives 0.50000012. In (vi) also check the rationalised form against the original at small n: at $n = 1$ both equal $(\sqrt{2}-1)/(\sqrt{3}-1) \approx 0.5658$.

Common slips

- Treating (iii) and (iv) alike: in (iii) the sine stays bounded while the denominator grows; in (iv) both top and bottom shrink to 0, and their ratio tends to 1.
- Reading $(\arctan n)^{-1}$ in (v) as the inverse function.
- Computing (vi) on a calculator straight from the original form at huge n: the subtraction of nearly equal numbers loses accuracy. Rationalise first.

Beyond STEP

At university, "$a_n \to L$" is defined precisely: for every $\varepsilon > 0$ there is an N such that $|a_n - L| < \varepsilon$ for all $n > N$. Every move above, such as the squeeze in (iii), taking reciprocals of a non-zero limit in (v) and pushing a limit through a continuous function in (vi), becomes a theorem proved from that definition in a first analysis course. Abbott's *Understanding Analysis* develops these ideas carefully and readably.

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